



Artificial intelligence (AI) and machine learning (ML) have a dominant role in providing predictive capability (Credit: <https://connectitude.com>)

fingerprint sensors, face scanners, and iris scanners, allow only the owner to access the car. Buttons and knobs are replaced with voice assistants. For instance, the car windows can function with touch screens and gesture controls.

Other smart features include self-assisted parking to provide safety against imminent collision, computerised diagnostics, GPS-based navigation, remote keyless entry using smartphones, lane detection while driving, and over-the-air (OTA) update systems.

Continuous improvements are taking place to assist the automotive industry in delivering the most advanced, safest, and comfortable

vehicles to its customers. While a few companies have been developing and testing self-driving prototypes on open roads, a number of manufacturers are already providing semi-autonomous capabilities, including automatic braking sensors, motorway lane sensors, and mapping technology. The future automobiles will provide drivers and passengers an extremely smooth journey and enable them to read, watch TV and work while just sitting inside their moving vehicles.

Evolution of automotive IT

Nicolas-Joseph Cugnot developed the first steam-powered automobile in 1769, and Karl Benz started the production in 1885. Later, Henry Ford created an automobile assembly line for mass-production. The Ford Model T was the first car to be mass-produced. Not long after Ford's Model T, automakers' focus shifted to outdoing one another on the basis of automobiles' speed, power, luxury, and performance. This led to increased use of IT in the automotive industry for creating cars that were safer, faster, more comfortable, economical, and environment friendly.

Today, IT is working towards the

goal of bringing self-driving smart vehicles in the automotive industry. **Some of the exciting new technologies enabling the creation of advanced future vehicles are improved software platform for autonomous ride-sharing service, predictive maintenance technology, self-balancing algorithms for personal pods, and new**

infrastructure designs for autonomous fleet vehicles.

In future, drivers will be able to control vehicles with just a wave of their hands using a 3D gesture technology system. They will be able to close and open windows by simply waving hands up or down, take an incoming call and control the airflow, temperature, etc, by pointing towards the screen of the infotainment system.

Predictive maintenance

Predictive vehicle maintenance determines when a vehicle needs to be serviced. The technology evaluates the vehicle's performance and informs the user about safety hazards. Artificial intelligence (AI) and machine learning (ML) have a dominant role in providing predictive capability by personalising a driver's driving experience and connecting vehicles to smartphones to take voice commands. Algorithms automate the process and update the optimal vehicular performance over longer periods of time. Remote servicing is also becoming a possibility. An example of predictive maintenance is Michelin's tire monitoring program that uses telematics and predictive analysis to monitor over-the-road tire condition in real time.

The collected data is fed into a computer to carry out real-time analysis of wear and tear of tires to identify the problem before it actually occurs or is close to it. The system maintains a continuous link between OEMs and customers, even as owners of vehicles change.

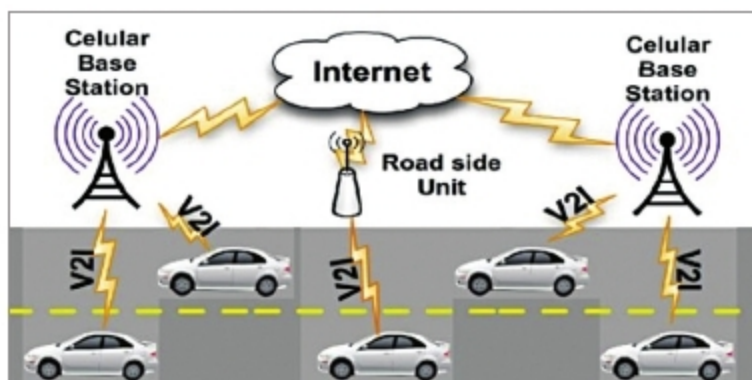
Communication with vehicles, infrastructure and networks

Imagine you are approaching a traffic light, and a vehicle suddenly jumps a red light. Though you fail to react to a sudden situation, your car provides a warning of a likely collision and applies brakes automatically in split seconds to avoid an accident. This is possible with vehicle-to-vehicle communication or V2V.

Wireless signals send information back and forth between vehicles about their location, speed, and direction to enable them to maintain safe distances from each other. V2V algorithms determine the best evasive measure required to preclude an



Vehicle-to-vehicle communication or V2V can prevent many accidents (Credit: <https://www.herrmanandherrman.com>)



Vehicle-to-infrastructure (V2I) technology enables many vehicles to share data with a range of devices (Credit: <https://www.researchgate.net>)